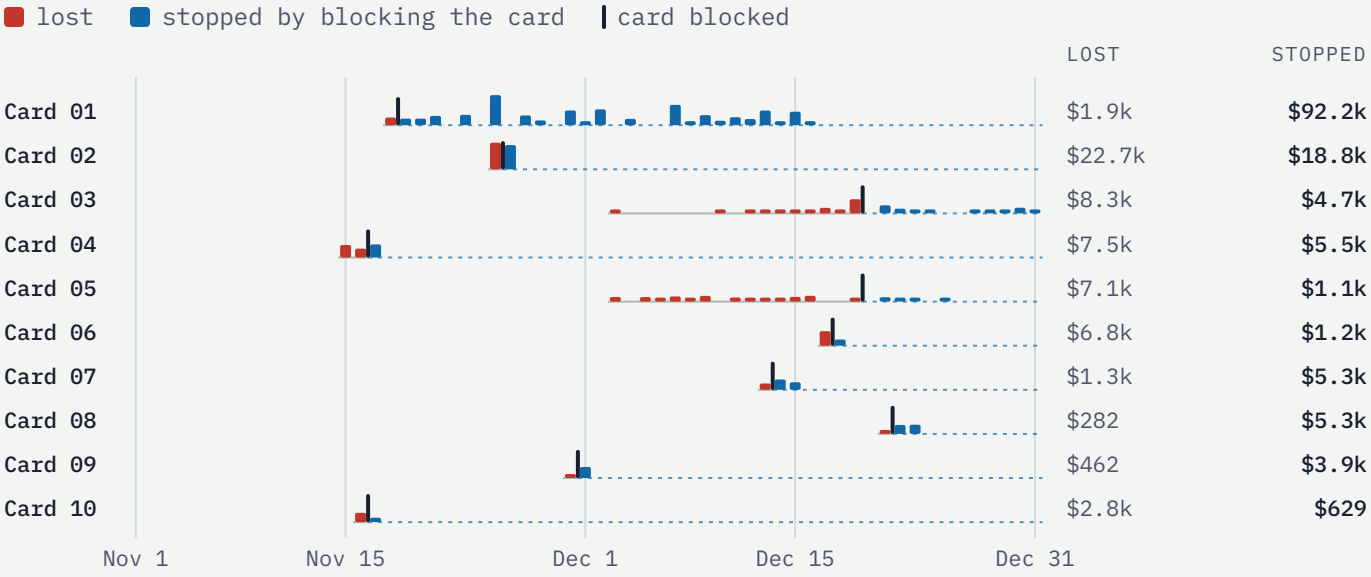
Stopping the card, not the transaction

A fraud score for a corporate purchase-card program: how well it holds up on months it never saw, and what it is worth when a confirmed alert blocks the card.

Jay Yu · UCLA Anderson, MS in Business Analytics

The ten largest card compromises in the out-of-time window, Nov–Dec 2010



Executive summary	2	6 · Card-stop value and alert briefs	9
1 · Where the fraud is	3	7 · Cutoff, staffing and uncertainty	11
2 · Data and cleaning	4	8 · Rollout and monitoring	13
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5 · Model selection and performance	7	B · Assumptions and reproducibility	17

Every figure in this report is generated by the fraudscore pipeline; see Appendix B.

TO Head of fraud operations, purchase-card program

FROM Jay Yu

RE Whether to deploy a transaction fraud score, and how to run it

Recommendation: deploy the score with a 3% review budget, block a card the same day its alert is confirmed, and run a 60-day shadow pilot before switching on real-time declines.

1. **It finds fraud on months it never saw.** On Nov–Dec 2010, held out from every modeling choice, the top 3% of scores contain **64.5%** of fraud transactions (Sep–Oct validation: **68.7%**). Optimized the same way, it is worth about twice the best single-variable rule: **$113k** against **$58.4k** net over two months.
2. **The value is in blocking the card, and speed decides it.** Fraud arrives in runs on a compromised card; **39%** of 2010 fraud dollars fell on a card's first fraud day. With the same alerts, blocking the next day stops **69%** of out-of-time fraud dollars, the same day **81%**, and declining in real time **88%**.
3. **It costs little to run.** At this program's volume the score raises **1.6–2.6** alerts a day, under a tenth of one analyst. Review, card reissue and customer friction together cost **2%** of the fraud dollars saved.
4. **The headline number is honest.** The usual random train/test split would have reported **84%** detection for a model that delivers **61%** on later months. Every choice here was made on earlier data only.

Value and uncertainty

Over the two out-of-time months the next-day policy nets **$113,057** at the agreed cost assumptions. Resampling the compromised cards and every assumption gives a 90% range of **$24k–$240k**, never below zero. The range is wide because only **16** cards were compromised in that window and ten cards hold **58%** of all fraud dollars.

Decision requested

Approve a 60-day shadow pilot: score live purchases, work alerts the same day, and log what a block would have stopped. Go live if the top 3% of scores catch at least 45% of fraud, the lower edge of what a 60-day sample can confirm (§8).

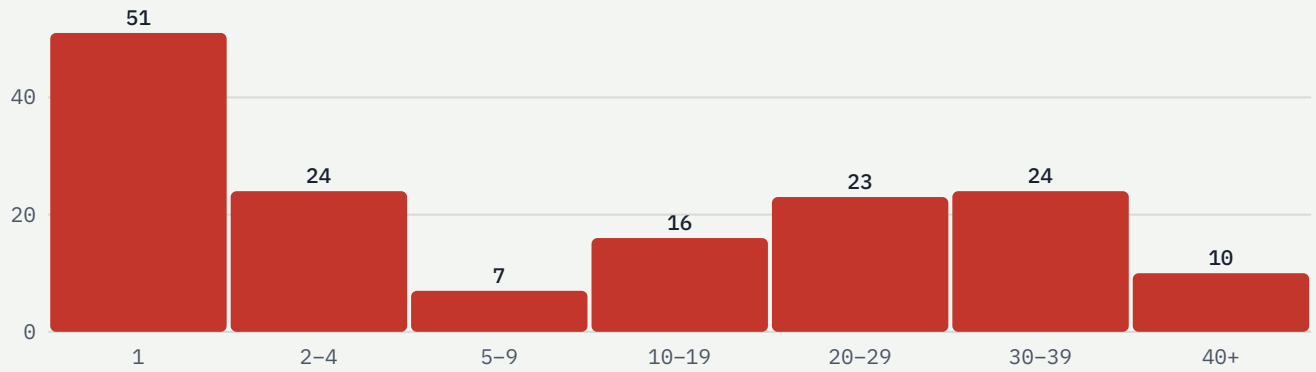
What it rests on

One program's 2010 transactions, fraud labels assumed complete, and cost inputs that are placeholders for the program's own figures (Appendix B).

Fraud is rare by transaction but concentrated by card: 155 cards carry all 2,283 fraud records.

97,805	2.3%	$2.49M	8
purchases on 1,643 cards in 2010	of purchases are fraud	of fraud over the year	frauds on the median compromised card

Figure 1.1 · Compromised cards by number of fraud transactions on the card, 2010



Count of the 155 cards with at least one fraud record.

What the shape means

A third of compromised cards (**51**) show a single fraud; the rest run for weeks, up to 69 transactions on one card. The ten costliest cards account for **58%** of fraud dollars, and **39%** of fraud dollars land on a card's first fraud day.

Three design choices follow

1. **Describe the card's recent behavior,** not the transaction alone: counts, spend and bursts over the last 0–30 days (§3).
2. **Test on later months,** because a random split puts the same card's burst on both sides (§4).
3. **Value an alert by the run it ends,** not the dollars it flags, and act fast (§6).

Two exclusions and three imputations make the file usable, and none of them looks at the fraud label.

The raw file has 98,161 records and ten fields: card, date, merchant number, description, state and ZIP, transaction type, amount and the fraud label. The full profile is in Appendix A.

Exclusions

RULE	RECORDS REMOVED	OF WHICH FRAUD
Non-purchase transaction type (A, D, Y)	355	0
Amount above $1,000,000	1	0

Only purchases (type P) are scored. The single $3.1M record is a data error, about 65 times the next largest amount ($47,900).

Imputation, most specific source first

FIELD	STEP	RECORDS
Merchant number	Missing or '0' before imputation	3,291
Merchant number	Most common Merchnum for the same description	1,166
Merchant number	New id created from description	2,125
Merchant state	Non-US code mapped to FOREIGN	203
Merchant state	Looked up from ZIP	57
Merchant state	Most common state for the same merchant	661
Merchant state	Most common state for the same description	1
Merchant ZIP	Most common ZIP for the same merchant	2,271
Merchant ZIP	Most common ZIP for the same description	28
Merchant ZIP	Most populated ZIP in the state	1,598
Merchant state	Left as UNK	312
Merchant ZIP	Left as UNK	469

Why this order

A missing merchant number is usually recoverable from its description, which is the same merchant. State comes from the ZIP when present, otherwise from the merchant's other records. ZIP comes from the merchant first and the state's most populated ZIP last, the weakest guess. Merchants that never had a number get a new one built from their description so they still group correctly.

Why it is safe

Cleaning uses identity and location fields only. Nothing here reads the fraud label, so using the whole file to resolve merchants cannot leak the outcome into the model. Statistics that do use the label, such as variable ranking, are computed on training rows only (§3–§5).

224 variables describe how each card, merchant and pairing behaved over the last 0–30 days, using only earlier records.

ENTITY	COUNT	TOTAL $	AVERAGE $	LARGEST $	THIS $ ÷ AVERAGE	DAYS SINCE LAST	BURST RATIO
Card	●	●	●	●	●	●	●
Merchant	●	●	●	●	●	●	●
Card × merchant	●	●	●	●	●	●	●
Card × merchant ZIP	●	●	●	●	●	●	●
Card × merchant state	●	●	●	●	●	●	●
Merchant × ZIP	●	●	●	●	●	●	●

Each statistic is computed for windows of 0, 1, 3, 7, 14 and 30 days. Burst ratio = transactions in the last 0–1 days ÷ the daily rate over 7, 14 or 30 days. Plus the amount itself and a round-dollar flag.

The rule that keeps them honest

A variable for a purchase may use that purchase and anything before it, never after. Same-day records count only if their record number is lower. A unit test rebuilds the table from a truncated file and checks that no earlier value changes.

Strongest variables in the final model

VARIABLE	MEANING	KS
card_total_3	Card: total spend, 3-day window	0.609
card_total_1	Card: total spend, 1-day window	0.607
card_count_1	Card: transactions, 1-day window	0.579
card_count_3	Card: transactions, 3-day window	0.573
card_total_0	Card: total spend, same day	0.566
card_total_7	Card: total spend, 7-day window	0.550
card_count_7	Card: transactions, 7-day window	0.533
card_count_0	Card: transactions, same day	0.524
card_total_14	Card: total spend, 14-day window	0.500
card_velocity_1_by_30	Card: activity 1-day window vs 30-day rate	0.452

A w-day window covers the purchase's day plus the previous w days. KS = largest gap between the fraud and non-fraud distributions, on Jan 15 – Aug 31 training rows. Top 10 of the 60 selected.

10

of the ten strongest variables describe the card itself: its recent count and spend.

60

variables kept, ranked by KS on training rows only, with near-duplicates (rank correlation > 0.95) removed.

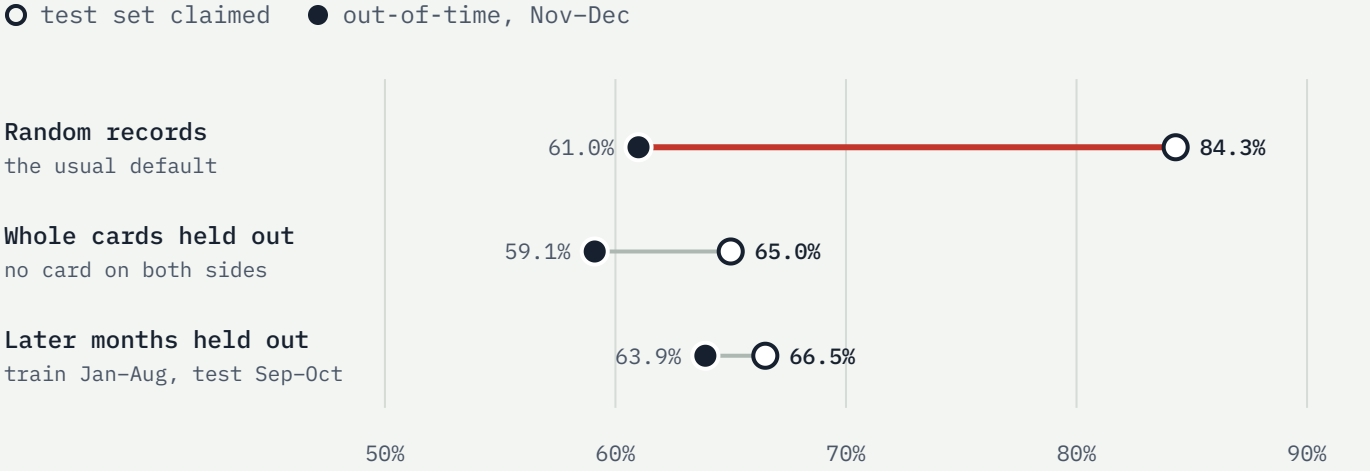
0

variables use the fraud label of any other record.

A random test split overstates detection by 23 points; testing on later months lands within 3 points of reality.

The same model was trained three ways and scored on its own test set and on the shared out-of-time window. A split is trustworthy if its test number predicts the out-of-time one.

Figure 4.1 · Detection rate in the top 3% of scores (FDR@3%), test set vs out-of-time, mean of 5 seeds



MODEL	SPLIT	TEST	OUT-OF-TIME	TEST − OOT	TEST FRAUDS ON CARDS WITH FRAUD IN TRAINING
Boosted trees	Random records	84.3%	61.0%	+23.3 pts	96%
	Whole cards held out	65.0%	59.1%	+5.9 pts	0%
	Later months held out	66.5%	63.9%	+2.6 pts	16%
Random forest	Random records	75.6%	51.9%	+23.7 pts	96%
	Whole cards held out	64.7%	53.1%	+11.6 pts	0%
	Later months held out	65.2%	55.3%	+9.9 pts	16%

Why the random split flatters

Under a random split, **96%** of test frauds sit on a card whose other frauds are in training. The model recognizes the burst it has already seen, which says nothing about the next compromised card.

What this report does instead

Hyperparameters are chosen on later months (§5), the cutoff on Sep–Oct (§7), and every headline number is out-of-time. A random forest on the same variables, shown for comparison, is also most overstated by the random split.

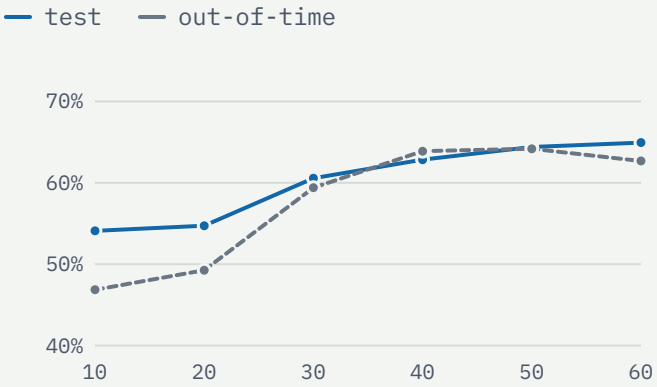
Boosted trees rank best, but the test windows cannot separate close settings, and tighter regularization does not help.

Each setting was trained on Jan 15 – Jun 30 and tested on Jul – Aug, then trained on Jan 15 – Aug 31 and tested on Sep – Oct. The pick is the best mean test score. Nov–Dec was recorded but never used to choose.

MODEL FAMILY	TRAIN	TEST (MEAN OF 2 FOLDS)	± 1 SE	OUT-OF-TIME
Boosted trees	**99.6%**	**65.4%**	**± 4.1**	**64.5%**
Random forest	87.8%	62.2%	± 3.8	58.8%
Logistic regression	66.3%	55.0%	± 4.5	43.6%

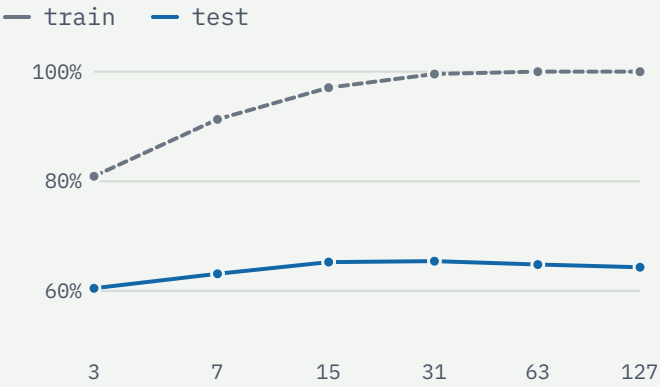
Best setting per family, FDR@3%. SE from resampling compromised cards in the test windows.

Figure 5.1 · More variables help



FDR@3% by number of variables kept.

Figure 5.2 · Bigger trees memorize, test stays flat



FDR@3% by leaves per tree, other settings fixed.

Noise sets the limit

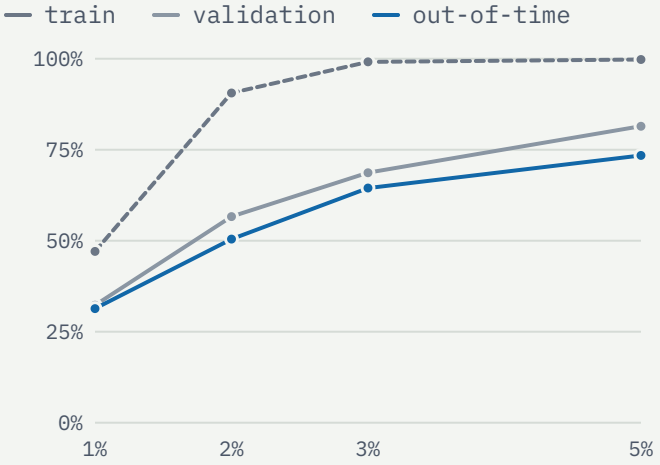
35 of **49** settings are within one standard error of the best. Each test window has about 45–50 compromised cards, so a 2-point difference is not evidence. The report treats the choice as "one good boosted model", not "the best one".

Overfitting, read correctly

Training detection near 100% looks alarming, but cutting trees to 3 leaves lowers it to 81% without raising test detection, and test falls below 7 leaves. The model memorizes training bursts without hurting its ranking of new cards. What does lower the number is time: Sep–Oct 69%, Nov–Dec 64%.

On Nov–Dec the model finds 64.5% of fraud in the top 3% of scores and 73% in the top 5%.

Figure 5.3 · Share of fraud found by review budget



WINDOW	FRAUDS	1%	2%	3%	5%
Train	1,413	47%	91%	99%	100%
Sep–Oct	415	32%	57%	69%	81%
Nov–Dec	**335**	**31%**	**50%**	**64%**	**73%**

FDR at each review budget. Training rows are scored in-sample.

Final model

Algorithm	histogram gradient boosting
Variables	60
Learning rate	0.03
Trees	300
Leaves per tree	31
Min. purchases per leaf	50
L2 regularization	1.0
Trained on	Jan 15 – Aug 31 2010

Against a rule an analyst could write

The strongest single variable, a card's spend over the last 3 days, used as a rule and tuned the same way, stops **$74.3k** of out-of-time fraud and nets **$58.4k**. The model stops **$144k** and nets **$113k** with the same next-day policy. The rule settles on a **2%** budget and **71** alerts; the model works **159** alerts and stops nearly twice the dollars, a trade worth making when an alert costs about $8 to review.

Stability

The score's distribution barely moved between training and Nov–Dec (population stability index **0.048**; below 0.1 is stable). The drop from validation to out-of-time comes from which cards were compromised, not from a shifted population.

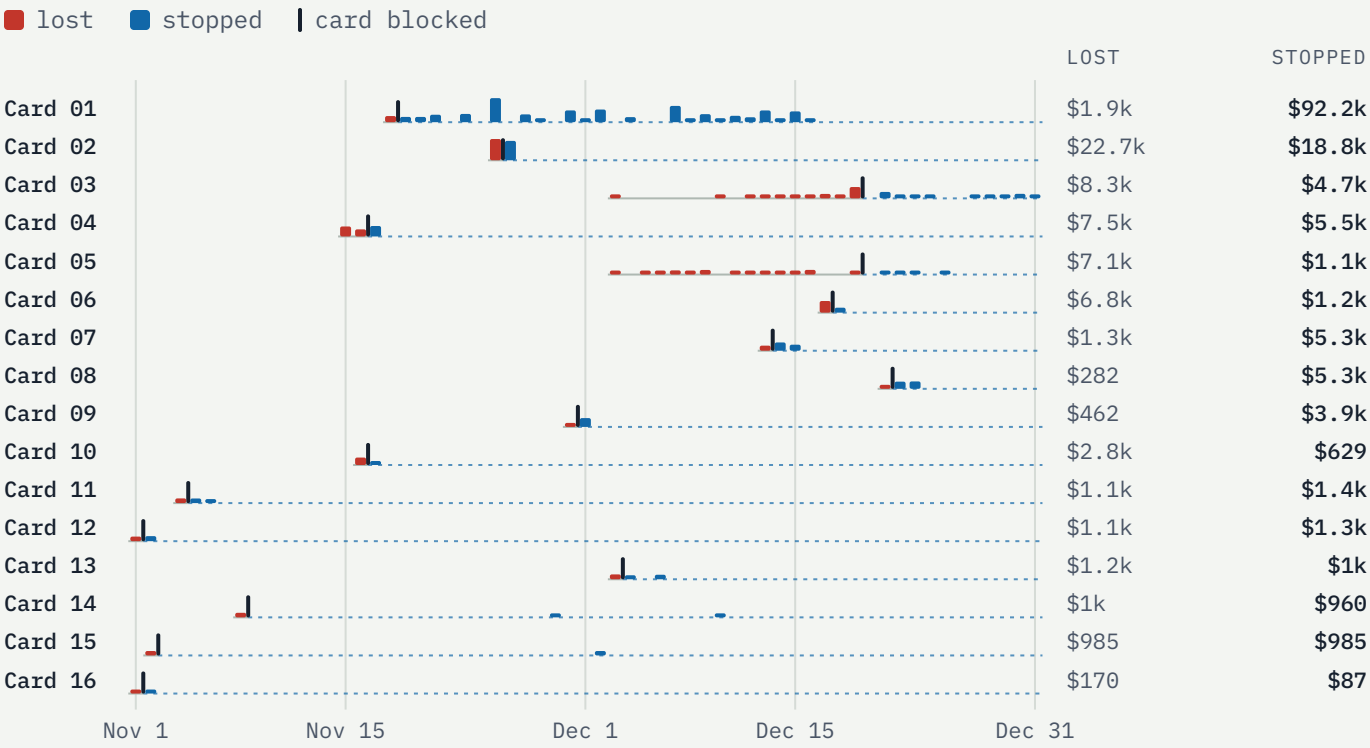
1.9×
net value of the model versus the best single-variable rule, Nov–Dec.

68.7%
validation FDR@3% (Sep–Oct), the number the cutoff was set on.

Blocking the card on its first confirmed alert stopped 69% of out-of-time fraud dollars. Acting the same day would have stopped 81%.

A confirmed alert is worth the rest of the run, not just the purchase it flags. Replay of Nov–Dec 2010 with the cutoff fixed on Sep–Oct; every alert worked, card blocked the next day.

Figure 6.1 · Every compromised card in the out-of-time window, Nov 1 – Dec 31



Bar height ∝ √(daily fraud $); cards renumbered by fraud dollars.

The first day decides most of the outcome

Card 01 was flagged on its first fraud day; blocking it the next morning stopped **$92,157** while **$1,888** was lost. Card 02 was also flagged on its first fraud day, but **$22,651** went through that same day, beyond the reach of a next-day block.

POLICY, SAME CUTOFF	STOPPED	SHARE	SEP–OCT	ALERTS	GOOD DECLINED
Block next day	$144,427	69%	39%	159	43
Block same day · recommended	**$169,991**	**81%**	**64%**	**100**	**59**
Decline in real time	$184,906	88%	82%	100	143

Good declined: legitimate purchases on a blocked card within the 5-day reissue lag, plus false real-time declines.

39%
of all 2010 fraud dollars fell on a card's first fraud day.

16 / 16
compromised cards in Nov–Dec were detected.

3 · $504
median frauds, and dollars, that got through before a card's first alert.

Every alert arrives with measured reasons and a calibrated fraud rate, so an analyst can confirm or clear it in minutes.

A score alone tells an analyst where to look, not what to check. Each alert gets a brief: the reasons it fired, how often alerts in its score band were fraud, the card's recent activity, and the action.

REVIEW

A-014 · C-09 · $420

Dec 13 2010 · score in the top 1.0% of purchases · 50% of Sep–Oct alerts in this band were fraud

1. **Card: purchases, same day:** 7 (typical 1; above 94% of purchases)
2. **Card: spend, 1-day window:** $1,180 (typical $320; above 77% of purchases)
3. **Card: average purchase, 7-day window:** $150 (typical $240; above 35% of purchases)

8 purchases on this card in the last 7 days, $1,200 in total; first purchase at this merchant.

Action: Confirm with the cardholder today; if the purchase is not recognized, block the card.

An actual out-of-time alert from the published sample. Card relabeled, dollar amounts rounded to $10.

How reasons are measured

Each variable is reset to its typical value, the median of non-fraud training purchases, and the drop in the score is that variable's contribution. Windows of the same statistic are grouped so a brief never lists "spend over 1 day" and "spend over 3 days" as two reasons. The method needs only the model's predictions, so it works for any model the program swaps in later.

Optional written summary

An LLM can rewrite a brief as two sentences (`fraudscore briefs --llm`, Claude Opus 5.5). It is kept only if every number in it appears in the brief; otherwise the template text is used. The guard is unit-tested with a stand-in client, so the pipeline never depends on the API.

89%
of Sep–Oct alerts in the urgent band (top 1% of scores) were fraud: call now.

50%
in the review band (1–3%): confirm today.

16 / 100
Nov–Dec alerts worked under same-day blocking were fraud; each one ended a card's run.

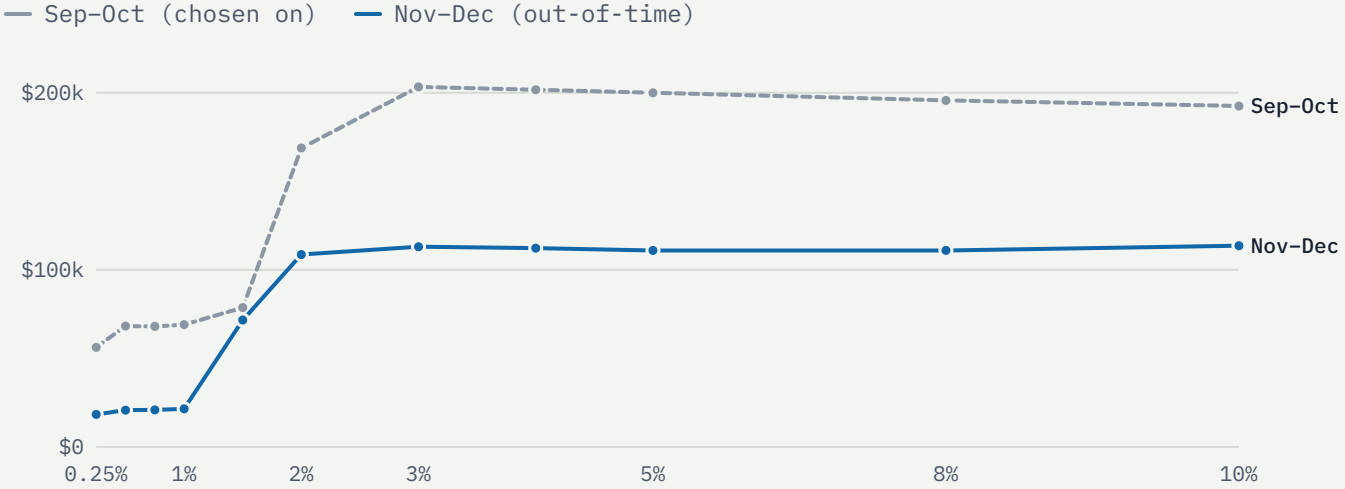
Top reason by outcome, every Nov–Dec alert

TOP REASON	TRUE ALERTS	FALSE ALERTS
Card: purchases	5	15
Card: spend	3	31
Card in this ZIP: spend	2	2
Card in this state: days since previous purchase	2	2
Merchant: this purchase vs. average	2	6
Card: largest purchase	1	12

Net value stops rising at a 3% review budget, which at this volume is 26 minutes of analyst time a day.

Net value = loss share × fraud dollars stopped − review time − card reissue − clearing false alerts − declined good purchases. The budget is chosen on Sep–Oct and applied unchanged to Nov–Dec.

Figure 7.1 · Net value by review budget, next-day blocking



Each window is two months. Steps appear where a large card gets caught earlier.

Why the curve is flat past 3%

Below 3% the score misses the early purchases of big runs, and value climbs in steps as each large card is caught earlier. Past 3% extra alerts are mostly false: going from 3% to 10% raises Nov–Dec alerts from **159** to **829** but fraud stopped only from **$144k** to **$156k**. Review costs are small (**2%** of savings), so the budget is set by where value stops rising, not by analyst cost.

If analyst time is capped

ALERTS WORKED PER DAY	NOV–DEC NET VALUE	UNWORKED ALERTS
2	$106,939	110
4	$109,257	63
8	$108,929	32
16	$113,172	12
unlimited	$113,057	0

First come, first served; the budget re-chosen on Sep–Oct for each cap.

Scaled to a program of about 1M purchases a year, the same 3% budget means roughly **0.7** analysts.

Only one assumption moves the answer, how much of unstopped fraud the program really loses, and the result stays positive across the range.

Figure 7.2 · Nov–Dec net value when one assumption moves to the end of its range

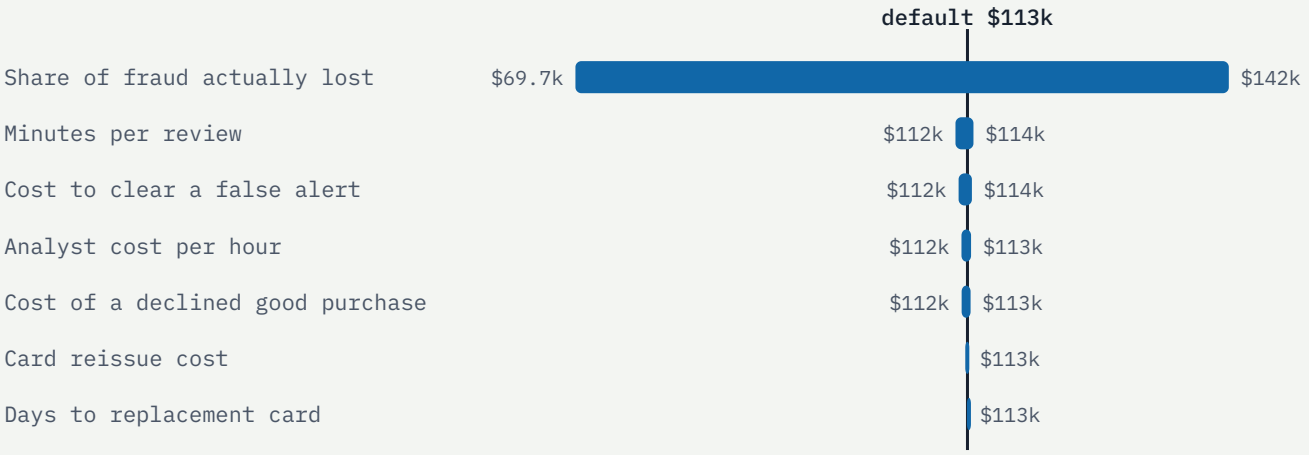
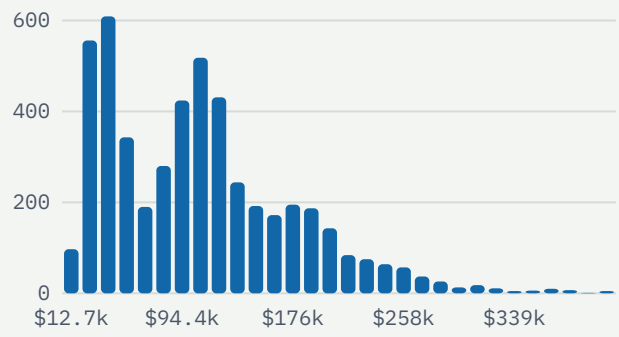


Figure 7.3 · Net value across 5,000 simulated outcomes



Assumptions drawn from triangular distributions over their ranges; compromised cards resampled with replacement.

Reading the range

The 90% range runs from **$24k** to **$240k** for two months, with a median of **$101k**, and no draw below zero. Almost all of the width comes from which cards happen to be compromised: holding every cost input at its default and resampling cards alone gives **$26.9k–$251k**. With **16** cards, one large compromise more or less moves the total by tens of thousands.

Why no annual number

Fraud dollars swing by month (Appendix A), so multiplying two months by six would mislead. The portable result is the share stopped: **69%–81%** of fraud dollars, depending on response speed.

A 60-day pilot can confirm the score works, not fine-tune it, so the go-live rule is set to what that sample can show.

Rollout in three steps

1. **Shadow, 60 days.** Score every purchase; analysts work the top 3% the same day but do not block. Log what a block would have stopped.
2. **Same-day blocking.** Switch on blocks for confirmed alerts. Keep a 10% random holdout of cards scored but not blocked as the comparison group.
3. **Real-time declines for the highest scores.** Only once the false-decline rate is known from steps 1–2.

How precise a pilot can be

COMPROMISED CARDS SEEN	ABOUT	FDR@3% ± 95%
16	**2 months**	**± 16 pts**
32	4 months	± 12 pts
48	6 months	± 9 pts
96	12 months	± 7 pts

Card-bootstrap standard error at Nov–Dec, scaled by √cards; months at the Nov–Dec rate of 8 compromised cards a month.

A 60-day pilot pins detection only to about ±**16 points**. The go-live rule is therefore "the top 3% catches at least 45%", which a working model clears and a broken one does not. Comparing two models needs a year of data, not a pilot.

What to watch, monthly

SIGNAL	NOW	ACT WHEN
Score stability (PSI vs training)	0.048	> 0.10 review, > 0.25 retrain
Most-shifted variable (PSI)	0.21	> 0.25 on a top-10 variable
Alert rate vs 3% budget	1.3%	outside 1–5% for two weeks
FDR@3% on confirmed fraud, rolling 90 days	64%	below 45%
Frauds before first alert, median	3	above 5

"Now" = Nov–Dec 2010 values. Most-shifted variable: card_state_days_since.

Controls

- **Labels arrive late.** Fraud is confirmed days to weeks after the purchase; measure FDR on a 30-day lag.
- **Retrain quarterly** on the latest 9 months, re-run the time-fold check, and keep the old model as challenger until the new one matches it.
- **Every alert carries measured reasons** and its band's fraud rate (§6), so analysts confirm in minutes and reviews are auditable.

The direction of the result is robust; its size rests on one year, one program and sixteen compromised cards.

Limitations

- **One program, one year.** Transactions are from 2010. Fraud tactics change; the time-fold design tests for drift over months, not years.
- **Small out-of-time sample.** 16 compromised cards in Nov–Dec. Shares are stable across validation and out-of-time in direction, not in size: next-day blocking stopped 39% in Sep–Oct and 69% in Nov–Dec.
- **Labels assumed complete.** Undetected fraud would be counted as good, which understates both fraud and the model's catch.
- **Blocking is assumed to work.** The replay counts every later fraud on a blocked card as stopped. A fraudster with several stolen cards would move on, so per-card value can be overstated.
- **Costs are placeholders.** Review time, analyst cost and friction costs are agreed planning values, not the program's figures. Only the loss share materially changes the conclusion (§7).

Next steps, in order of value

1. **Same-day operating model.** The largest lever is process, not modeling: alerts worked within hours lift the share stopped from 69% to 81%.
2. **Cardholder self-confirmation** by text message for review-band alerts, to cut analyst time further.
3. **Merchant-side detection.** Compromises often share a merchant; a merchant-level score could block cards before their first fraud.
4. **Real-time scoring service** with the same point-in-time variables, for declines at the highest scores.
5. **Longer history.** Two to three years of data to measure year-over-year drift and tighten the value range.

WHAT WOULD CHANGE THE RECOMMENDATION

If the program recovers most fraud losses through chargebacks, the case weakens but holds: at a loss share of 0.3 the next-day policy still nets **$40.8k** over Nov–Dec. If the pilot's FDR@3% comes in under 45%, stop and revisit the variables before going live.

Appendix A · Data quality report

Raw file before exclusions: 98,161 records, 10 fields, one row per card transaction, Jan 1 – Dec 31 2010.

Field summary

FIELD	TYPE	POPULATED	UNIQUE	MIN	MEDIAN	MAX	MOST COMMON SHARE
Recnum	numeric	100.0%	98,161	1.00	49,081.00	98,161.00	–
Cardnum	categorical	100.0%	1,645	–	–	–	1.2%
Date	date	100.0%	365	2010-01-01	–	2010-12-31	–
Merchnum	categorical	96.5%	13,091	–	–	–	10.0%
Merch description	categorical	100.0%	13,126	–	–	–	1.7%
Merch state	categorical	98.8%	227	–	–	–	12.6%
Merch zip	categorical	95.2%	4,567	–	–	–	12.9%
Transtype	categorical	100.0%	4	–	–	–	99.6%
Amount	numeric	100.0%	35,452	0.01	137.20	3,102,045.53	–
Fraud	numeric	100.0%	2	0.00	0.00	1.00	–

Field notes

- **Recnum** is a row identifier in date order; it breaks ties within a day.
- **Cardnum** has no missing values; 1,645 cards.
- **Merchnum** is blank on 3.5% of records and '0' on 237 more; recovered from descriptions where possible (§2).
- **Merch state** includes numeric codes for foreign merchants, grouped as FOREIGN.
- **Merch zip** is the least complete location field; imputed from merchant, then state.
- **Transtype** is almost entirely P (purchase); the others carry no fraud and are excluded.
- **Amount** runs from $0.01 to $3.1M; the $3.1M record is an error.
- **Fraud** is the label: 1 for fraud.

Transaction types

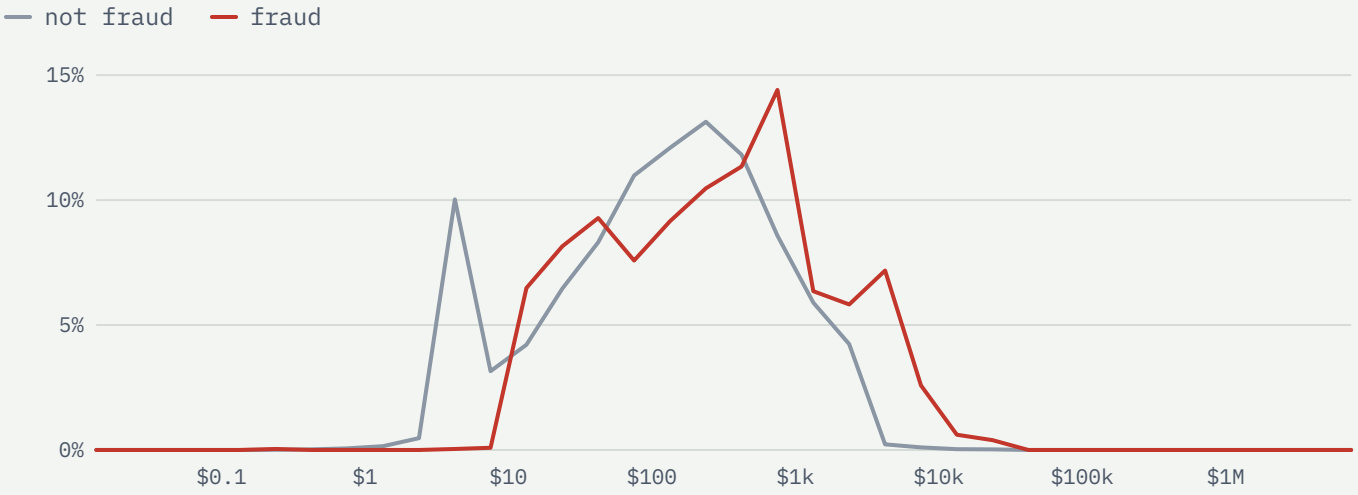
TYPE	RECORDS
P	97,806
A	181
D	173
Y	1

Top merchant states

STATE	RECORDS
TN	12,206
VA	7,982
CA	6,919
IL	6,593
MD	5,490
GA	5,115

Appendix A · Distributions

Figure A.1 · Purchase amount, share of records per bin (log scale)



Bins evenly spaced on a log scale from $0.01 to $10M. Fraud skews to mid-sized purchases.

Figure A.2 · Records per month

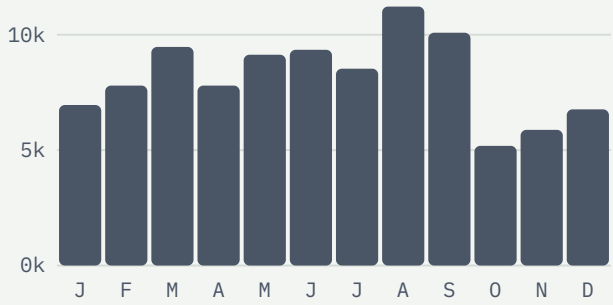
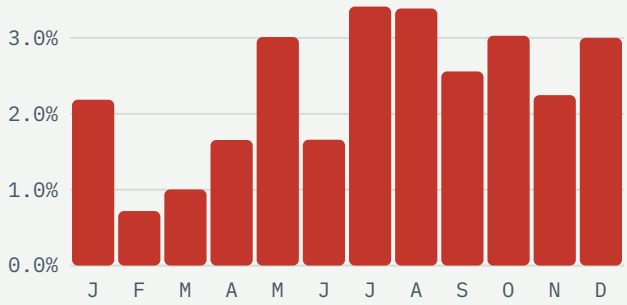


Figure A.3 · Fraud rate per month



October volume is well below other months, and monthly fraud rates move between 0.7% and 3.4%. Both are reasons to judge the model on later months rather than a random sample.

Appendix B · Assumptions and reproducibility

Cost assumptions

ASSUMPTION	USED	RANGE TESTED
Share of fraud actually lost	80%	50% – 100%
Minutes per review	10 min	5 min – 20 min
Analyst cost per hour	$50	$35 – $75
Card reissue cost	$15	$10 – $30
Cost to clear a false alert	$5	$0 – $15
Cost of a declined good purchase	$10	$2 – $25
Days to replacement card	5 days	3 days – 7 days

Planning values agreed for this analysis, not sourced benchmarks. Replace with the program's own figures; the decision tool recomputes every result.

Reproduce every number

`fraudscore run`	cleaning, variables, validation audit
`fraudscore tune`	time-fold search, final model
`fraudscore decide`	cutoff, staffing, response policy, uncertainty
`fraudscore briefs`	reason codes and analyst briefs
`fraudscore export`	aggregates for site, report, deck, README
`fraudscore synth`	synthetic data to run it all without the dataset
`scripts/build.sh`	PDFs and figures

`fraudscore all` runs everything in about eight minutes on a laptop. Unit tests cover cleaning order, point-in-time variables, split boundaries, FDR, PSI, replay policies, cost arithmetic, reason codes and the LLM grounding guard; CI also runs the whole pipeline on synthetic data.

Definitions

- **FDR@k%**: share of fraud transactions found among the k% highest-scored transactions.
- **Out-of-time (OOT)**: Nov 1 – Dec 31 2010, never used for any choice.
- **Review budget**: share of purchases raised as alerts, set as a score cutoff on Sep–Oct.
- **Prevented**: fraud dollars declined because the card was already blocked (or, in real time, declined at the alert).

Data and authorship

About 98,000 labeled purchases from one purchase-card program in 2010. The transaction file is not redistributed; this report and the site publish only aggregates and a relabeled, rounded alert sample. Analysis, code and report: Jay Yu.